



## Linking prominence, and other cues, to the processing of focus across languages

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A key function of focus is to highlight what is said in relation to what could be said, or alternatives to the focus. This plays a crucial role in conveying implied meaning in discourse. Psycholinguistic research has established that focus-marking has processing effects consistent with this contrastive focus function in a range of languages, including drawing attention to contrastive referents, priming alternatives and suppressing non-alternatives in lexical activation, and strengthening memory for mentioned alternatives. These studies have used primary focus markers for the language, largely contrastive prosodic prominence. However, research has shown that an astonishing range of cues can signal focus, including phonological prominence, syntactic construction and word order, grammatical role, verb semantics and the presence of gesture, as well as fine phonetic detail at the segmental, suprasegmental and multimodal levels. Weighting of these cues differs between languages, speakers and with the discourse context. I review my own group's and related research on focus processing and cues to focus across languages from the perspective of the listener. I offer a view of how these might link, based on focus as an attentional tool, drawing attention to intended focus alternatives, based on probabilistic expectations about prominence.

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## 1. Introduction

When we speak, much of what is meant is implied rather than spoken, and it is essential that listeners understand this for the coherence of the discourse and to pragmatically interpret what is said as intended. Focus-marking plays a key role in this implied meaning, including in implying alternatives to the focused word. For example, if a speaker says “The TEACHER finished the green tea”, this implies that there is a question under discussion in the discourse of who finished the green tea, and that the speaker intends the listener to evaluate alternatives to *teacher* who could have drunk the tea in the context. These implied meanings are crucial to the interpretation of the utterance.

In recent years, psycholinguistic evidence has grown for processing effects of focus consistent with this function of implying alternatives in a range of languages. This work has principally looked at effects of contrastive prosodic prominence (indicated by capitals in the example above), although some has looked at other primary markers of focus in the language, including syntactic clefting. However, both within and across languages there are a very wide range of potential cues to focus, including phonetic and phonological prominence, syntactic construction and word order, grammatical role, verbal semantics and multimodal cues. We are as yet far from understanding how these interact and are evaluated by listeners and how that relates to how listeners process focus, and how this might differ across languages and individuals.

Below, I briefly set out what focus is and its functions in discourse interpretation, before reviewing what prosodic prominence is and how it is expressed across languages. I then discuss research from my group, and related work, looking at psycholinguistic evidence for the function of focus as implying alternatives in speech processing, in English and other languages. I then review work from our group, and others, looking at the weighting of different cues to focus across languages, in particular prosodic prominence, syntactic clefting and grammatical role; as well as other cues from production studies. Finally, this is brought together to sketch a proposal of how we might link cues to focus, and processing of focus alternatives, across languages.

## 2. Focus alternatives and discourse interpretation

There is a very wide literature on focus, and differing views on how it should be defined (e.g., see Féry & Ishihara, 2016; Kruijff-Korbayová & Steedman, 2003; Vallduví, 2016). In this paper, I am primarily concerned with the concept of *focus* as captured in Krifka’s informal definition:

Focus indicates the presence of alternatives that are relevant for the interpretation of linguistic expressions. (Krifka, 2008 p. 247)

This conception of focus, as implying the presence of alternatives to the focus, is the central claim of the Alternative Semantics framework (Rooth, 1992). Under Rooth’s theory, focus-marking

introduces, along with the ordinary semantic value of a proposition, a focus semantic value, which is a set of alternatives which can replace the focused element in that proposition.

- (1) [The TEACHER]<sub>F</sub> finished the green tea.
- (2)  $\lambda x$  [finished( $x$ , green tea)] |  $x \in E$ , where  $E$  is a set of alternatives of the same semantic type
- (3) The *pupil* finished the green tea. / The *principal* finished the green tea. / The *taylor* finished the green tea.

For example, in (1), *teacher* is focused, marked by a contrastive prominence (indicated by capitals, see further next section). This implies a set of alternatives to the focus, which are of the same semantic type, and which could replace *teacher* in the proposition (2). Examples are given in (3). Importantly, these alternatives are relevant for the interpretation of the utterance, that is, part of the intended meaning of the utterance is the implication of these alternatives and how they are relevant. The listener therefore needs to resolve the intended alternative set to fully interpret the utterance. Note that Rooth's theory is silent on the means of focus-marking, though in English this is assumed to include contrastive pitch accenting.

As can be seen in (3), the focus semantic value can include a wide set with both alternatives likely to be plausible in the context, like *pupil* and *principal*, and those that are not, like *taylor*. In Rooth's theory, the alternative set, in terms of the focus semantic value, is highly permissive, constrained only by semantic type, here the type of individuals. It is then a pragmatic process to constrain the alternative set to relevant, contextually plausible alternatives.

To understand how the intended alternatives can be resolved, we need to situate focus within information structure. During a discourse, interlocutors build up a common ground of information that they believe to be established with the other speaker(s) (Clark, 1996; Stalnaker, 1978). The information structure of an utterance is how each entity, predication, and so on is marked as to how it refers back to, alters, and/or updates the existing discourse model (Calhoun, 2010; Krifka, 2008; Vallduví, 2016). The alternative set should have an antecedent in the context, or the alternative set should be able to be accommodated, so resolving the alternative set follows similar processes to other kinds of antecedent resolution (Rooth, 2016; Vallduví, 2016).

- (4) Who finished the green tea?
- (5) All the food was gone and there was an empty cup on the table. Emily wondered what had happened...

For example, the focus may update a *question-under-discussion* (QUD) in the context (Beaver & Clark, 2008; Roberts, 1996). Under a QUD approach, each assertion in a conversation answers a question-under-discussion either explicitly or implicitly available in the context, thereby forming an important part of conversational dynamics and coherence. If (1) was said in the context of (4), the focus directly updates the QUD established by the explicit question. However, the

QUD is frequently not explicit like this, but implicit. If (1) was said in the context of (5), the listener is invited to accommodate a QUD like (4), in particular from the noting of an *empty cup*. Likewise, the intended alternatives may be mentioned in the context, either directly, as a superordinate set (e.g., *food and drink*) in the example above, or they need to be accommodated by the listener. Focus-marking can also be used to reactivate an open proposition that is no longer salient in the conversation, e.g., if the interlocutors had talked about *the green tea* much earlier in a conversation.

(6) I think the caretaker finished the green tea.

One specific use of the QUD updating function is to correct or confirm information in the common ground. For example, if (1) was said in response to (6), it would have the specific function of correcting the referent in the proposition in (2) established in the common ground. In these various ways, focus-marking plays an important role in regulating the salience of both QUDs and relevant entities in relation to them.

(7) I know principal finished the cake, but what about the teacher?

(8) The [TEACHER]<sub>F</sub> finished the [GREEN TEA]<sub>F</sub>.

While the QUD and the focus frequently align, they are not equivalent, as there can be alternatives to the QUD as well. In response to (7), (8) contains foci on both *teacher* and *green tea*, as there is an implied set of alternatives to both. However, they are of different kinds, *teacher* in this context is a contrastive topic or theme, so the alternative set is effectively of the QUDs that the speaker may be intending to update, e.g., *what did the principal finish?*, *what did the teacher finish?*, *what did the caretaker finish?*, etc. (Büring, 2003; Calhoun, 2010; Steedman, 2014). In this way, focus-marking plays a role in discourse management and coherence: the existence of an alternative set of QUDs indicates that the current answer is incomplete (Büring, 2003; Krifka, 2008). This may be resolved in the preceding context (as in the example in (7)/(8)), but it may also indicate upcoming topics. So focus-marking can play a role in frame-setting or delimitation of subsequent discourse.

In many languages, including English, subjects are prototypical topics and objects sit in the default position for focus, whether this is defined by syntactic position (the focus is the most deeply embedded argument) or prosodic position (the focus is the final strong element in an intonation phrase) (Calhoun, 2010; Chafe, 1976; Y. Chen et al., 2016; Halliday, 1967; Ladd, 2008). This means that it is less likely that subjects will be focus-marked than objects or other arguments (i.e., an example like (1) is actually rather uncommon). When subjects are foci, this is marked, either as a contrastive topic (in an example like (7)), or where the subject is accessible in the context (i.e., not in ‘out-of-the-blue’ contexts). On the other hand, when the object is focus-marked, this can be ambiguous between narrow focus on the object itself, and broad focus on the whole verb phrase or even clause (Calhoun, 2010; Ladd, 2008).

Finally, focus-marking is crucial to the interpretation of certain focus-sensitive operators, such as *only*, *even* and *also*, in ways that can affect the truth-conditional semantics of the utterance (Beaver & Clark, 2008; Krifka, 2008; Vallduví, 2016).

(9) The teacher only [FINISHED]<sub>F</sub> the green tea.

(10) The teacher only finished the [GREEN TEA]<sub>F</sub>.

The semantics of *only* specifies that only the focus-marked alternative is true within a contextually relevant set of alternatives. For example, (9) implies that some alternative to *finished* in the proposition is not true, e.g., that *the teacher just started the green tea*; whereas (10) implies that some alternative to *green tea* is not true, e.g., *the teacher finished the coffee*. While the exact relationship to the alternative set differs between these operators, utterances containing them cannot be fully interpreted without resolving the alternative set.

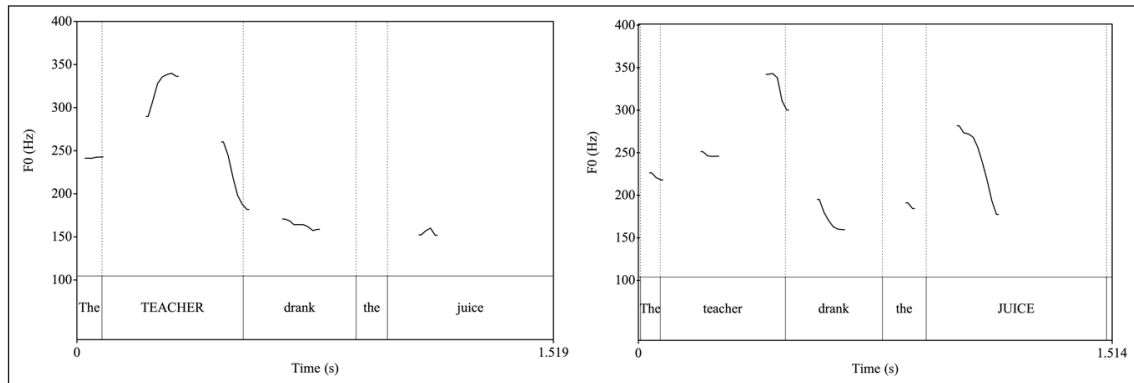
In summary, a central function of focus-marking is to indicate the presence of alternatives to the focus-marked constituent. Resolving the intended alternative set is important to the interpretation of the utterance in the discourse, including in updating and making salient entities in the common ground, and updating and managing questions-under-discussion.

### 3. Prosodic prominence

In the examples above the capitalised words are described as having contrastive prosodic prominence. Prosodic prominence plays a clear role as a primary cue to focus, and its interpretive effects just described, across many if not most languages (Kügler & Calhoun, 2020). However, what is meant by prosodic prominence is debated (Ladd & Arvaniti, 2023). Here, we take *prosodic prominence* to be the prosodic means that are used in a language to make one (or more) element (usually a prosodic word) in an intonational phrase stand out or be highlighted in relation to the rest of that phrase. Within this, we distinguish between *phonological* and *phonetic prominence*. Phonological prominence refers to cues to prominence that form part of the prosodic phonology of the language, or structural means that mark one prosodic word as the strongest within its phrase. There are at least three broad types of phonological prominence: stress-based, register-based and phrase-based (Kügler & Calhoun, 2020). Phonetic prominence refers to gradient means to adjust the prominence of an element which is phonologically prominent, typically including higher pitch, greater pitch movement, lengthening, and increased loudness (Fletcher, 2010; Turk, 2011). It can sometimes be difficult to draw a precise boundary between these, but the distinction is still useful.

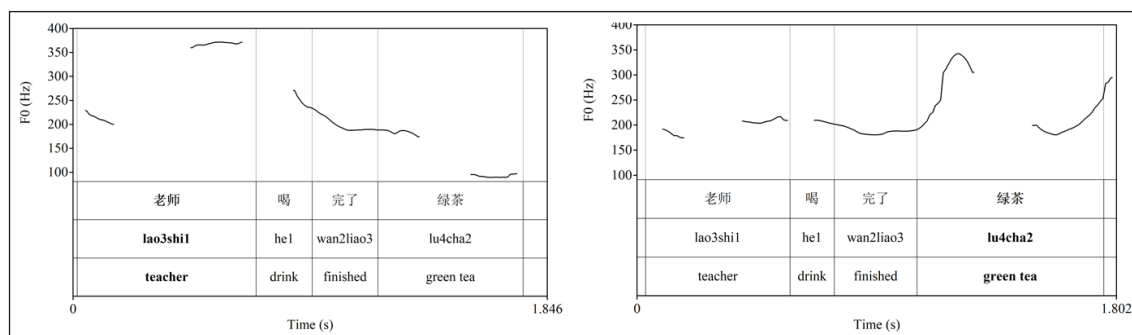
Finally, we refer to *contrastive (prosodic) prominence*. This is perhaps the trickiest of the concepts, as it blurs the line in what is otherwise a clean distinction between *focus*, a semantic/pragmatic concept, and *prominence*, a phonetic/phonological one. Nonetheless, studies on focus processing frequently use particularly emphatic or strong prominence to cue

focus which is unambiguously contrastive (re the previous section). Whether such contrastive prominence is a different categorical type to noncontrastive prominence remains disputed, and likely differs between languages. For English, in Calhoun (2009) and (2010), I have argued it is more fruitful to think of this as a gradient distinction linked to the likelihood of a contrastive interpretation.



**Figure 1:** Example of a contrastive pitch accent (caps) on the subject (left) and object (right) in English (from the stimuli in Calhoun et al., 2025, see section 4).

English, which is the most widely studied language in the psycholinguistics literature, has stress-based phonological prominence, in common with many European and other languages (Calhoun, 2010; Kügler & Calhoun, 2020; Ladd, 2008). The focused word usually carries a pitch accent associated with the most prominent, or stressed syllable in the intonational phrase, called the *nuclear accent*. **Figure 1** shows stimuli from Calhoun et al. (2025) (see section 4). As can be seen, the contrastively prominent words are marked with phonetic cues to prominence, including high/rising pitch associated with the stress syllable of the accented word, lengthening and intensity. However, prominence is also phonological, meaning it is also cued by the structural position of the accent. In **Figure 1**, focus on the subject (*teacher*) is marked by a very high, rising (L + H\* in ToBI) accent on the subject, and low, flat pitch afterward. Such a pattern is generally needed to signal focus on the subject. With focus on the object (*juice*), there is also a large, rising accent on the object; however, there is still an accent on *teacher*, and a rising phrase tone (H-) at the end of *teacher* that is higher than the accent on *juice*. Such prenuclear accents do not generally mark focus, although a particularly strong prenuclear accent may (e.g., (8)) (see Calhoun, 2010; Kügler & Calhoun, 2020). A strong ‘contrastive’ accent on the object is also not necessarily needed to mark focus. As the nuclear accent is structural, it may also be cued by rhythmic cues in the preceding prosodic structure in the intonation phrase (Calhoun, 2010; Ladd, 2008; Ladd & Arvaniti, 2023).



**Figure 2:** Example of contrastive prominence (bold) on subject (left) and object (right) in Mandarin Chinese (from the stimuli in Yan et al., 2023, see section 4).

In Mandarin Chinese, local pitch movements are determined by the lexical tone on the syllable, so there are no pitch accents. Mandarin has a register-based prominence system, where prominence is achieved through the expansion of the pitch register of the focused word; as well as lengthening and intensity on the prominent word (Y. Chen & Gussenhoven, 2008; K  gler & Calhoun, 2020). In Mandarin, though not in all register-based prominence systems, there is also postprominence register compression. Tonal targets are also implemented much more precisely, i.e., closer to citation form, with less coarticulation effects, in prominent positions compared to postprominent (focal) positions (Y. Chen, 2010). In **Figure 2**, The tonal movements on the subject are much larger and better defined when it is focused (left) than not (right) and vice versa for the object. There are similar asymmetries in the marking of prominence on subjects versus objects based on position to English, with strong compression postfocally but not prefocally.

Almost all the studies discussed below involve languages with stress- or register-based prosodic prominence systems. The third type of phonological prominence identified in the literature to date is phrase-based prominence (see K  gler & Calhoun, 2020), e.g., Seoul Korean, where focus is marked through the insertion of a phrase tone before the focused constituent, higher scaling of phrase initial tones in the focused constituent and optionally ‘de-phrasing’ afterward (e.g., Jeon & Nolan, 2017; Jun, 2011; Jun et al., 2007). That is, the phonological expression of prominence is primarily related to manipulation of phrasing. Mixtures of stress-based, register-based and phrase-based systems are also found, for more discussion see K  gler and Calhoun (2020).

#### 4. Focus alternatives and processing

Given the role of focus and the alternative set in discourse interpretation described above, it follows that listeners should be highly attentive to focus-marking cues and that focus plays important roles in speech processing. Indeed, it has long been shown that focus-marked words are attended to more and remembered better. Early phoneme-monitoring experiments in English

and Dutch (Akker & Cutler, 2003; Cutler, 1976; Cutler et al., 1997; Cutler & Fodor, 1979), and more recently in Mandarin Chinese (Ip & Cutler, 2021), showed listeners attend more to focused words marked with contrastive prominence, or where the preceding intonation contour would lead the listener to expect prominence. In written tasks with syntactic clefts, focus-marked words were recalled more quickly and accurately in immediate and longer-term processing in English and Korean (Birch et al., 2000; Birch & Garnsey, 1995; Gernsbacher & Jescheniak, 1995; Kember et al., 2021; Sanford et al., 2009). Kember et al. (2021) showed the effectiveness of different focus-marking cues differs by language, with clefting more effective than prosodic prominence in Korean, while these were equally effective in English. Focus can guide parsing and interpretation under ambiguity in English (Carlson & Potter, 2022; Schafer et al., 2000; Sedivy, 2002). Focus, marked with a variety of means, affects reading times and regressions during silent reading in English (Birch & Rayner, 2010; Hoeks et al., 2023; Lowder & Gordon, 2015; Ward & Sturt, 2007).

Crucially for our purposes, there is now a range of psycholinguistic evidence that focus-marking causes listeners to generate a set of alternatives to the focus-marked constituent, and that alternatives play important roles in the processing of utterances (Gotzner & Spalek, 2019). Eye-tracking studies in English, Dutch and German have shown that contrastive accenting leads listeners to expect a contrastive referent (Braun et al., 2019; Braun & Biezma, 2019; Dahan et al., 2002; Dennison, 2010; Ito & Speer, 2008; Kim et al., 2015; Kurumada et al., 2014; Watson et al., 2008; Weber et al., 2006). For example, Ito and Speer (2008) showed that when following instructions to decorate a Christmas tree with multiple ornaments varying in colour and shape (e.g., *blue angel*, *green ball*), listeners anticipatorily looked at, for example, the *green ball*, if after placing a *blue ball* they heard “now hang the GREEN...”, and took longer to fixate on the correct referent if this was actually, for example, a *green angel*. This work has also shown, using the visual world paradigm, that listeners pay selective attention to alternatives arising from inferences related to focus-marking (Braun et al., 2019; Dennison, 2010; Kim et al., 2015; Kurumada et al., 2014).

Another line of studies have used the cross-modal priming paradigm to show that focus-marking, particularly with contrastive prosodic prominence, selectively activates alternatives to the focused word (Braun & Tagliapietra, 2010; Calhoun et al., 2025; Husband & Ferreira, 2016; Yan et al., 2023; Yan & Calhoun, 2019). It has been long established using semantic priming experiments that on hearing a word, semantically associated words are activated (as well as phonologically associated words) (Collins & Loftus, 1975; Quillian, 1967). However, this activation was found to be less robust in sentence contexts (Norris et al., 2006). Braun and Tagliapietra (2010) were the first to show that this is partly related to focus-marking: in their study on words in object position in Dutch, semantic associates that were contrastive alternatives were primed only when the word was contrastively accented, while non-contrastive associates were weakly primed regardless of focus-marking.

In Yan, Calhoun and Warren (2023), we found similar results for words in subject position in Mandarin Chinese.

(11) *Prime:*

- (a) [老师]<sub>F</sub> 喝 完 了 绿茶。  
lao3shi1 he1 wan2 liao3 lu4cha2  
teacher drink finish PAST green tea  
‘[The teacher]<sub>F</sub> finished the green tea.’ ([S]<sub>F</sub> VO, *subject focus-marked*)
- (b) 老师 喝 完 了 [绿茶]<sub>F</sub>。  
lao3shi1 he1 wan2 liao3 lu4cha2  
teacher drink finish PAST green tea  
‘The teacher finished [the green tea]<sub>F</sub>.’ (SV[O]<sub>F</sub>, *subject not focus-marked*)

(12) *Target:*

- (a) 老师 lao3shi1 ‘teacher’ (identical)  
(b) 校长 xiao4zhang3 ‘principal’ (contrastive alternative)  
(c) 小学 xiao4xue2 ‘primary school’ (non-contrastive associate)  
(d) 太空 tai4kong1 ‘space’ (unrelated)

In the experiment, participants heard a sentence, like (11), where the subject prime was either focus-marked (a) with prosodic prominence (via pitch register expansion, indicated by underline) or not (b). They then saw a target which was identical to the subject, a contrastive alternative, a noncontrastive semantic associate or an unrelated control (12). They had to respond as quickly as possible if the target was a word (in filler trials, there were non-words which were nonsensical character combinations). It was found that participants were faster to respond to both the identical and alternative targets in the focus-marked than not focus-marked condition, while focus-marking did not significantly affect response times for the noncontrastive and unrelated targets. This is consistent with prosodic focus-marking selectively activating an alternative set to the focused word. In another study with a similar design, we showed that a combination of prosodic and syntactic focus-marking (via clefting) primed alternatives, but not syntactic focus-marking on its own in Mandarin (Yan & Calhoun, 2019).

For English, we ran experiments with a similar design, looking at priming of both alternatives to the subject and object, and including a further target type, unrelated alternatives, e.g., *builder* in the example above (Calhoun et al., 2025). We found a somewhat different pattern of results to what we found for Mandarin (and Braun & Tagliapietra, 2010, for Dutch, although their experiments had a partially different design). Participants were faster to respond to the alternative targets than unrelated controls in both the focus-marked and not focus-marked condition. However, they were only faster to respond to the noncontrastive associate targets than unrelated controls in the not focus-marked condition. This, however, matches the results

of Husband and Ferreira (2016) for English. As they propose, this is consistent with a selection mechanism, where initially all semantically associated words are activated, but focus-marking causes noncontrastive associates to be suppressed as not relevant to the alternative set. These experiments all indicate focus-marking selectively activates alternatives to the focused word, but whether the mechanism is best conceptualised as one of activating alternatives, or suppressing noncontrastive associates, may differ by language, or more dynamically given specific details of the task and stimuli (see Calhoun et al., 2025; Husband & Ferreira, 2016).

We also reported a novel finding, that unrelated alternatives to the prime object were primed regardless of focus-marking, e.g., an alternative like *eggplant* could potentially replace *green tea* in the prime sentence (9), but is not semantically related to *green tea* and unlikely to be contextually appropriate. This is consistent with Rooth's theory that, initially at least, a wide, permissive set of alternatives may be activated, that are then narrowed to contextually plausible alternatives as a pragmatic process (see Calhoun et al., 2025; Gotzner, 2017).

In the second experiment, targets were related to the subject. Over the whole experiment, there was an unexpected finding that all target types were primed in comparison to the unrelated controls, regardless of focus-marking; an effect that was attributed to learning over the course of the experiment. An exploratory analysis of the first quartile of the experiment, however, showed a similar pattern as for objects: related alternatives were primed relative to unrelated controls across focus-marking conditions, while noncontrastive associates were only primed in the not focus-marking condition. This indicates focus-marking selectively activates focus alternatives for both subjects and objects. However, unrelated alternatives were not primed in either focus condition. We attribute this to the differing role of subjects and objects in discourse, with subjects typically given or accessible in the context, even if they are contrastive. This finding suggests that activation of a very wide alternative set may depend on context (see further Calhoun et al., 2025).

In these experiments, activation of alternatives was shown for focused words in sentences presented out of context. Other experiments have looked at whether the activation of alternatives is consistent with key roles alternatives are claimed to play in the activation and tracking of referents in discourse contexts (see section 2). Drawing on an earlier experiment by Gotzner, Wartenburger and Spalek (2016) on German, we used the probe recognition paradigm to look at how focus-marking affects the activation of mentioned versus unmentioned alternatives in discourse contexts in Samoan (Calhoun et al., 2023).

- (13) *Context:* Sa fa'atau e le mālō ma lona to'alua meaai: 'o le pai, 'o le falaoa ma le keke.  
'The visitor and her husband bought some food: a pie, bread and cake.'
- (14) *Continuation:* Sa u'u e le tamāloa le falaoa ma le pai.  
'The husband carried the bread and the pie.'

(15) Critical sentence:

'O [le mālō]<sub>F</sub> sa 'aia le keke.

PRES DET visitor PAST eat-ES DET cake

'It was [the cake]<sub>F</sub> that the visitor ate.' (object focus-marked)

'O [le keke]<sub>F</sub> sa 'ai e le mālō.

PRES DET cake PAST eat ERG DET visitor

'It was [the visitor]<sub>F</sub> who ate the cake.' (object not focus-marked)

(16) Probe:

(a) le keke 'the cake' (*identical*)

(b) le falaoa 'the bread' (mentioned alternative)

(c) le ēsi 'the papaya' (unmentioned alternative)

(d) le kolisi 'the college' (*unrelated control*)

In the experiment, participants first heard a context (13) which introduced two alternative sets, e.g., *people* and *food*. They then heard a continuation sentence which repeated alternatives from each set (14). In the critical sentence, the object was either focus-marked or not (15). Focus-marking was using a syntactic cleft-like structure, which our previous research showed is the primary marker of focus in Samoan (Calhoun, 2015; Calhoun et al., 2021). The focus is fronted, preceded by the particle 'o. This is also the usual position for nuclear prosodic prominence, but our previous work has shown the prosodic cue is much less perceptually important than the clefting one in Samoan (Calhoun et al., 2021, see section 6). Participants then saw a probe (16), and had to respond as quickly as possible whether it was in the preceding story. There was no effect of focus-marking on response times for the identical probes, probably due to floor effects. Participants were numerically slower to correctly verify mentioned alternatives when the object word was focus-marked than not, although this did not reach significance, and significantly slower to correctly reject unmentioned alternatives (compared to unrelated controls) in the focus-marked condition. This was similar to the findings in Gotzner et al. (2016) for German (for focus marked with focus particles *only* or *even*, with non-contrastive accents). As they argue, this shows that focus-marking activates a whole alternative set, which in short-term processing makes it harder to distinguish between mentioned and unmentioned alternatives, as they are all part of the activated alternative set.

Other studies suggest, however, that in delayed processing (from a few minutes to a day), focus-marking makes listeners more accurate in correctly recalling mentioned alternatives and distinguishing them from unmentioned alternatives.

- (17) (a) Both the British and the French biologists had been searching Malaysia and Indonesia for the endangered monkeys.  
 (b) Finally, the BRITISH/British spotted one of the monkeys in INDONESIA/Indonesia and planted a radio tag on it.

In Fraundorf, Watson and Benjamin (2010), participants heard a context sentence like (17) (a) introducing two alternative sets, and then a critical sentence in which the repeated member of one or both of those sets was contrastively accented or not (17)(b). Participants were more accurate at recalling the correct alternative if it had been contrastively accented both immediately after hearing a long block of such narratives and the following day (see also Fraundorf et al., 2013). Using narratives similar to (13)–(15) in German, Spalek, Gotzner and Wartenburger (2014) found participants had more accurate recall of mentioned alternatives when the word was focus-marked with a focus particle. Tjuka, Nguyen and Spalek (2020) showed similar findings in a delayed recall task with similar stimuli in Vietnamese, with prosodic focus-marking, though this was only significant for female speakers. In Calhoun et al. (2023), in Samoan, when listeners heard blocks of narratives like (11)–(13), participants were faster to correctly recognise mentioned alternatives when the object was focused (compared to unrelated controls). In the second experiment in Yan et al. (2023), participants heard blocks of sentences like (11) in Mandarin Chinese, and then had to say if the targets had been mentioned. In this case, they were slower to correctly reject alternatives as not having been in the previous block if the prime word was focus-marked (although this only held in the first quartile of the experiment due to learning effects). This is likely because the sentences were produced out of context, so participants were not encouraged to build a detailed representation of the alternative set from mentioned alternatives in the discourse.

In summary, there is now substantial psycholinguistic evidence for focus-marking selectively activating alternatives to the focus-marked word, either through activation of alternatives or suppression of non-alternatives. This is found in lexical priming experiments, and also in experiments showing consequent effects of alternative activation in discourse processing, including anticipation of contrastive referents in a visual scene, and immediate activation and longer-term memory for alternatives in short discourses. These experiments have primarily used contrastive prominence, although equivalent effects have been shown with other primary markers of focus for the language, e.g., with syntactic clefting in Samoan.

## 5. Cues to focus across languages

As just shown, there is considerable psycholinguistic evidence that focus-marked constituents are attended to more in processing, and that focus-marking triggers the activation of an alternative set, which is resolved in relation to the discourse context. These studies have primarily been on Germanic languages (English, Dutch and German); however, there is some evidence for these focus processing effects in other languages including Mandarin Chinese, Samoan, Korean and Vietnamese. In most of these studies, the focus marker was contrastive prosodic prominence, although our Samoan study showed equivalent effects with syntactic clefting (Calhoun et al., 2023). (Some of the English and German studies used clefts or focus particles in combination

with prosodic prominence, or used written materials where a nuclear accent on the focused word in the cleft was likely because of implicit prosody). That is, the position of focus within critical sentences was reasonably unambiguous, using a primary focus-marker for that language. However, as will be discussed in this section, there are multiple potential cues to focus, including phonetic and phonological prominence, syntactic structure, grammatical role, semantic verb type and multimodal cues like gesture. There is still much to learn about how these cues interact and how listeners determine the position of focus, and how this affects focus processing, especially when different cues conflict.

We have explored these questions in a series of studies in different languages looking at how listeners weight cues to the position of focus, comparing contrastive prosodic prominence and syntactic clefting, including when they conflict (Calhoun et al., 2021; Yan et al., 2022; Yan & Calhoun, 2022; see also Arnhold, 2021; S.-H. Chen et al., 2012).

- (18) *Context:* The family had a few pets. Different members of the family gave each pet their dinner.
- (19) *Question choice:*
- (a) Who fed the mouse? (*subject question, SQ*)
  - (b) What did the sister feed? (*object question, OQ*)
- (20) *Critical sentence:*
- (a) The SISTER fed the mouse. (*canonS*)
  - (b) The sister fed the MOUSE. (*canonO*)
  - (c) It was the SISTER who fed the mouse. (*ScleftS*)
  - (d) It was the sister who fed the MOUSE. (*ScleftO*)
  - (e) It was the MOUSE that the sister fed. (*OcleftO*)
  - (f) It was the mouse that the SISTER fed. (*OcleftS*)

In Calhoun et al. (2021), with English listeners, participants first saw a context like (18), which introduced two different alternative sets. They were then shown two questions which could be asked about that context (19). They then heard a critical sentence (20) which was either in canonical word order ((a),(b)), a subject cleft ((c),(d)) or an object cleft ((e),(f)). Within each of these, there was a contrastive accent on either the subject ((a),(c),(f)) or the object ((b),(d),(e)). They then saw the questions again, and had to choose which question they thought was most likely to have been asked given the way the speaker answered. The questions were about either subject (SQ)(19)(a) or the object (OQ)(19)(b) of the critical sentence. It was found that in the canonical order sentences, as expected, the focus was perceived to be in the position of the contrastive accent (canonS = SQ, canonO = OQ). In the cleft sentences, when the accent fell on the cleft, participants judged the focus to be in the cleft (ScleftS = SQ and OcleftO = OQ). These conditions were not significantly different to the canonical sentences, showing the effect of the

clefing and prominence cues were not additive where these agreed. Where the cues to focus did not agree, participants were more likely to follow the clefing cue (ScleftO = SQ and OcleftS = OQ), although the preference for each question was weaker than when the cues agreed.

We ran a parallel experiment in Samoan (Calhoun et al., 2021), with the same six conditions in the critical sentence: canonO, canonS, ScleftS, ScleftO, OcleftO and OcleftS. The clefts were ‘o-fronting constructions (as in (15)). The results showed that in Samoan the clefing cue was weighted much more strongly. In the cleft sentences, participants judged the focus to be in the cleft, and the accent position had no significant effect (ScleftS, ScleftO = SQ and OcleftO, OcleftS = OQ). In the canonical word order sentences, participants overall weakly preferred the object question, with no significant difference based on accent position, however, there was a lot of variability between listeners in this condition, with some preferring the accented word, some the subject and some the object; we suspect this may be related to factors including relative dominance in Samoan and English (as all participants were bilingual), although unfortunately the design of the experiment did not allow this to be investigated further.

For Mandarin, we used a somewhat different task, acceptability judgment (Yan et al., 2022).

- (21) (a) 是 [女儿]<sub>F</sub> 喝完 的 果汁。  
 shi4 nu3er2 he1wan2le0 de0 guo3zhi1  
 SHI daughter finish DE juice  
 ‘It was [the daughter]<sub>F</sub> who finished the juice.’ (ScleftS)
- (b) 女儿 是 喝完 的 [果汁]<sub>F</sub>。  
 nu3er2 shi4 he1wan2le0 de0 guo3zhi1  
 daughter SHI finish DE juice  
 ‘It was [the juice]<sub>F</sub> that the daughter finished.’ (OcleftO)

The clefts were *SHI...DE* clefts (Paul & Whitman, 2008). As in English, the cleft head usually carries nuclear prominence, and is analysed as being in focus. However, unlike English and Samoan, the order of the subject and object does not change. Rather, for a subject cleft, SHI immediately precedes the subject, as in 21(a), while in object clefts it is before the verb, as in 21(b). In both cases, DE precedes the object, which also indicates past tense. As in English and Samoan, it is also possible to put prominence on a different constituent, so there are competing syntactic and prosodic cues to focus within the utterance, e.g., prominence on 果汁 ‘juice’ in 21(a) (ScleftO) or on 女儿 ‘daughter’ (OcleftS) in 21(b).

Participants first saw a context, similar to (18). They then heard a question, which was either SQ or OQ (like (19)), then the critical sentence, which was again one of canonO, canonS, ScleftS, ScleftO, OcleftO or OcleftS. They were then asked to rate how appropriate the answer was to the question on a 1–7 point scale. As expected, and parallel to the English results, for canonical word order answers, question answer pairs where the focus in the question matched the prominence in the answer were rated highly (canonS = SQ, canonO = OQ). When the prominence was in the

cleft, and this matched the focus, these were also rated highly (ScleftS = SQ and OcleftO = OQ), although ratings were slightly lower for OcleftO clefts. However, opposite to English, when the cues to focus did not agree, listeners followed the prominence cue, rating answers with matching prominence highly (ScleftO = OQ and OcleftS = SQ), although ratings were lower than when the cues agreed.

Finally, we ran an experiment with the same question-choice task as Calhoun et al. (2021) with L1 Mandarin L2 speakers of English (Yan & Calhoun, 2022) (see also Yan et al., 2022 for results using the appropriateness rating task with L2 learners). In canonical order sentences, listeners perceived the focus to be on the prominent word, similar to L1 Mandarin and L1 English. For the cleft sentences, the results were more similar the L1 English learners, i.e., they perceived the focus to be in the cleft, with prominence not in the cleft weakening but not reversing this preference. Therefore, the L2 listeners seemed to have reversed the weighting of prominence and clefting cues from their L1. However, judgements where the prominence cue matched the focus were more definite than for the L1 English listeners, suggesting the L2 listeners were more affected by prominence cues than the L1 English listeners, potentially due to their L1 Mandarin influence.

These studies show that, in a metalinguistic judgment task, both prosodic prominence and syntactic clefting affect listeners' judgments about where the focus is, with the weighting of these cues differing between English, Samoan, Mandarin and for L1 Mandarin L2 English speakers. All of these studies looked at focus on both subjects and objects, finding little difference in responses between them in each language. However, in focus processing tasks, we have found the grammatical role (subject/object) substantially affects responses (see also the finding in Section 4 that unrelated alternatives to focused words are primed for objects but not subjects in English).

- (22) *Context:* The couple helped the traveller and his friend find a hotel. They were both very thankful.
- (23) *Connecting question:* Can you tell me more?
- (24) *Critical sentence:* The traveller thanked the woman.  
(as *canonS*, *canonO*, *ScleftS*, *ScleftO*, *OcleftS*, *OcleftO*)
- (25) *False alternative:*
  - (a) Did the friend thank the woman? (SQ)
  - (b) Did the traveller thank the man (OQ)

Yan and Calhoun (2020) report parallel studies in English and Mandarin using a false alternatives task to investigate focus processing (with L1 speakers of each language) (see also Ayers, 1996; S.-H. Chen et al., 2012). Participants saw a context like (22) which introduced two alternative sets. They then heard a connecting question (23). They then heard the critical sentence, which

included one member of each alternative set (e.g., *traveller*, *woman*) in one of six syntactic/prosodic types (24) as in the studies reported above (Calhoun et al., 2021; Yan et al., 2022). They then saw a question about that sentence in which, in critical trials, either the subject or object was false, and they had to answer ‘no’ as quickly as possible. It was predicted participants would be faster when the focus-marking in the critical sentence matched the false information in the question, and slower when it did not.

For subjects, in both languages, the pattern of results broadly matched that for the focus judgment task. In Mandarin, participants were faster to respond to the false subject questions when the prominence was on the subject than on the object, with no difference between syntactic conditions (canonS, ScleftS, OcleftS). In English, participants were faster when the accent was on the subject than object in canonical order (canonS v. canonO), but not for subject clefts (ScleftS vs. ScleftS) or object clefts (OcleftS vs. OcleftO). Responses to object clefts were also slower than the other two types (OcleftS vs. canonS/ScleftS). This is consistent with the focus judgment task: in both languages prominence is an effective cue to focus, and matching clefts do not enhance this cue. However, mismatching clefts weaken cues to focus in English but not Mandarin, showing clefting is a weighted more highly as a cue to focus in English.

For objects, however, the pattern was quite different. In both languages participants were not faster with prominence on the object than on the subject in canonical order or for subject clefts. There was also no difference for object clefts in English, though OcleftS was slower than OcleftO and canonS in Mandarin, which may be due to the relative infrequency of this construction in Mandarin. Otherwise, there were no significant effects of syntactic construction in either language.

In Yan and Calhoun (2020), we attribute this difference between subjects and objects to a default object focus bias (see also Calhoun et al., 2025). This is reflected in the asymmetry in their discourse properties (prototypical topics versus foci, see Section 2), and how they are marked prosodically in both languages (see Section 3). In processing, the effect of this is that listeners tend to process objects as focused even if they are not overtly focus-marked, as was found here. Similar effects have been found in processing of subjects versus objects in focus-sensitive ellipsis structures (Carlson et al., 2009; Harris, Jesse & Carlson, 2018). Interestingly, this effect of grammatical role was only found in this processing task, not the judgment task. This suggests listeners are aware that the object is not focus-marked, but nonetheless the default bias to process it as such persists.

From production studies, we can see that other kinds of syntactic or semantic bias may affect the perception of focus. For instance, in Calhoun, La Cruz and Olssen (2018), we looked at the interaction of focus marking and semantic verb type in intransitives in Spanish. In Spanish, as in English, the default position for the nuclear accent is on the final object in broad focus (e.g., see Zubizarreta, 1998). However, for intransitives, this is often not the case. For instance, it is claimed that when the verb is unergative (e.g., *brincar* ‘to jump’), the nuclear accent is on that

word; whereas when it is unaccusative (e.g., *salir* ‘to leave’), the accent often shifts to the subject, and there may also be a word order reversal (in English, this accent shift happens but not the word reversal), as (27) (Ladd, 2008; Zubizarreta & Nava, 2011).

- (26) (a) ¿Qué pasó?  
           what happen-PAST  
           ‘What happened?’ (broad focus question)
- (b) ¿Quién brincó?  
           who leave-PAST  
           ‘Who jumped?’ (subject focus question)
- (c) El joven brincó, ¿es verdad?  
           DET child leave-PAST BE-PRES right  
           ‘The young man jumped, right?’ (contrastive subject question)
- (27) (a) La chica BRINCÓ,  
           DET girl jump-PAST  
           ‘The girl JUMPED.’ (Unergative verb)
- (b) La CHICA salió. / Salió la CHICA.  
           DET girl leave-PAST leave-PAST DET girl  
           ‘The GIRL left.’ (Unaccusative verb)

In the experiment, Venezuelan Spanish speakers described a picture showing an intransitive action in response to a question intended to elicit broad focus, subject focus or contrastive focus on the subject (26). The picture prompt included the verb the speaker should use, which was either unergative or unaccusative. Results showed that in broad focus, speakers were more likely to have the nuclear accent on the subject with unaccusative verbs, as expected (with or without word reversal). In response to subject focus questions, the accent was also more likely to be on the subject, but less so for unergative verbs; while for contrastive focus, the accent was most likely to be on the subject, with no difference by verb type. While this was a production study, if we think about it from a perception point of view, this shows that the weight of an accent on the subject in an intransitive sentence in Spanish as a cue to focus varies depending on the semantic verb type (i.e., it is more ambiguous if the verb is unaccusative). As discussed in Calhoun et al. (2018), underlining these effects may be inherent differences in the information structure properties of unergative versus unaccusative verbs, but nonetheless, this implies that such semantic factors affect the perception of focus.

Research has also shown that cues to focus can be multimodal, including gesture. Perception studies have shown that constituents which are gestured, especially with head nods or eyebrow raises, are more likely to be interpreted as focused (e.g., Krahmer et al., 2002 in Dutch; and Prieto et al., 2015 in Catalan). We have shown results consistent with this in production for Turkish.

In Türk and Calhoun (2024), we looked at the co-occurrence and synchronisation of information structure, prosodic phrasing and gesture in spontaneous narrations. Results showed that focused phrases were more likely to be gestured than topic phrases and background (68% versus 27% and 5% respectively), especially with iconic gestures (see also Ambrazaitis & House, 2017 in Swedish; and Fung & Mok, 2018 in Cantonese). But the results suggest not just the presence, but also the timing of gesture matters. In general, we found that prosodic phrases and gesture phrases are synchronised; however, this differed by the type of phrase (prenuclear, nuclear, postnuclear). For nuclear phrases, which is the usual location of focus in Turkish, the synchronisation of prosodic and gesture phrase onsets and offsets was nearly perfect (mean difference close to 0 ms). This implies that a tighter synchronisation between prosodic and gestural structure can be a cue to focus. Further, while prosodic and gestural structure are generally synchronised across languages, the details of this synchronisation differ depending on the prosodic phonology of the language, so these cues are language-specific (e.g., Türk & Calhoun, 2023).

The same is true for prominence itself. Prosodic prominence is signalled by a wide range of fine phonetic cues at the suprasegmental and segmental levels, which differ across languages, but broadly involve phonetic cues appropriate to the phonological expression of prominence in the language, as well as more careful articulation of the prominent syllable or larger prosodic constituent (e.g., see Cho, 2016; Kügler & Calhoun, 2020). For example, Jang, Kim and Cho (2018) showed that there was less nasalisation of vowels next to nasal consonants under focus in Korean, consistent with less coarticulation under focus. Mitterer, Kim and Cho (2024) showed this kind of segmental effect can affect perception, listeners were sensitive to the (lack of) segmental reduction of an auxiliary verb as a cue to focus in German, even in the absence of F0 information. Prominence is crucially also relative, i.e., the perception of phonetic cues to prominence in one part of an intonational phrase depends on the prominence of other parts of the same phrase (Calhoun, 2010; Ladd & Arvaniti, 2023). Importantly for our purposes, such phonetic cues to focus are gradient, not simply present or absent, and this interacts with the perception and processing of focus, in ways that are far from being fully understood. For example, in an early eye-tracking study, Watson, Tanenhaus and Gunlogson (2008) showed, as expected, that contrastive L + H\* accents led listeners to expect a contrastive reference (see Section 4). However, with ‘neutral’ H\* accents listeners were equally likely to look at the contrastive or (noncontrastive) new referent in the visual scene, which the authors describe as showing that the interpretative domains of the two accent types overlap, i.e., ‘non-contrastive’ H\* accents can also be compatible with a contrastive focus interpretation.

As we can see, there are a wide range of factors that can signal focus. While we expect the types of factors, including phonetic and phonological prominence, syntactic construction, semantic verb type, and multimodal cues, to be similar across languages, there are important cross-linguistic differences in how these are weighted and how they interact, which are far from

being well understood. This paper has concentrated on the perception and processing of cues to focus, while the majority of studies in the field have looked at the production of focus (e.g., see references in Kügler & Calhoun, 2020). Perception and processing studies have only looked at a small subset of cues, such as those reviewed above, in a narrow range of languages. It is safe to assume that the relationship between production and perception of cues to focus is not straightforward. More studies, on both production and perception, need to account for the interaction of multiple cues, in a wider range of languages.

Further, the majority of studies in this area have looked at first language speakers of each language, assuming a (near-)monolingual language context. We know that language contexts are often more complex than this, and multilingual speakers are likely to show different weightings of cues to focus reflecting influence of their different languages. We showed this above for focus perception in L1 Mandarin L2 English speakers. For heritage language speakers, and in language contact situations, interesting new patterns of the functional use of prosodic and syntactic cues to focus can emerge (Calhoun et al., 2016, 2024; Delais-Roussarie et al., 2015). Finally, research in recent years has shown differences in the perception of different cues to prominence and focus between individuals within a language community, relating to factors including neurodiversity, empathy and pragmatic skill (Bishop, 2016; Grice et al., 2016; Orrico et al., 2023).

## 6. Expected prominence and pragmatic contrast

We have seen that focus has important functions in discourse processing, particularly in implying the presence of alternatives that are relevant to the interpretation of utterances, and there is substantial psycholinguistic evidence to support these functions. These studies have largely looked at primary focus markers in the language, particularly contrastive prosodic prominence. However, as we have just outlined, there are a very wide range of cues that can signal focus, including phonological prominence, the syntactic construction/word order, semantic verb type and the presence of gestures, as well as fine-grained phonetic and multimodal cues in the realisation of prominence. The weighting of these cues differs between languages and speakers. Added to this is the effect of the discourse context itself, as discussed in Section 2, focus is part of discourse management, including QUD updating; therefore, focus on a particular constituent can be more or less expected given the preceding context including the preceding QUD. Bringing these together, we need to understand how different listeners evaluate and use these cues to both locate the focus and process it as intended.

We have seen that prosodic prominence and syntactic clefting are weighted differently as cues to focus by speakers of different languages, including English, Mandarin and Samoan. Other research has shown that multiple cues affect the perception of prominence, and this differs across languages. For example, in another experiment reported in Calhoun et al. (2021) (see section 5), listeners heard the stimuli (similar to (18)) out of context and had to rate which word they thought

was the most prominent. In the Samoan experiment, listeners were strongly affected by syntactic focus, being about 15–20% more likely to choose the nuclear accented words in a syntactic cleft as the most prominent than an accented word in the main clause in a cleft construction (ScleftS/OcleftO versus ScleftO/OcleftS). Participants also choose the verb as the most prominent word around 19% of the time, likely as this is the default nuclear accent position in Samoan, even though it was never intended to be nuclear accented. In the English experiment, listeners choices were also significantly affected by the syntactic construction, but this effect was much weaker, with listeners choosing the intended nuclear accented word as the most prominent over 95% of the time in all conditions. Effects of syntactic, semantic and informativity factors have been found to interact with phonetic and phonological factors in the perception of prominence across multiple languages (Baumann & Winter, 2018 on German; Bishop et al., 2020 on English; Cole et al., 2010 on English, Cole et al., 2019, on English, French and Spanish; Luchkina & Cole, 2021 on Russian; Turnbull et al., 2017 on English; Vainio & Järvikivi, 2006 on Finnish).

What I would like to suggest is that we can think of focus as an attentional tool, or cognitive instruction to process the focused constituent in a certain way. This is along the lines of this formulation from Calhoun (2009 p. 61, bold added):

“[the] pragmatic effects of contrastiveness arise because speakers use **probabilistic expectations** about the prosody/information structure interface precisely in order **to get ...pragmatic effects**. Speakers indicate through the **manipulation of expected prominence** that their utterance **means more** that its propositional content/information structure suggests. This **draws attention to the contrastiveness** of the [contrastive focus] **and** therefore **intended implicatures** arising from the presupposition of its alternative set.”

That is, prominence arises from probabilistic expectations about how a particular utterance is said, arising from cues from the utterance itself, from the phonetic and phonological cues to focus that are there, but also the words, their semantic value, their syntax, multimodal cues etc.; and also from the discourse context, including multimodal cues. These create an impression of whether a particular constituent is more prominent than expected, or whether the speaker is drawing attention to it, in a probabilistic way. We can then relate this to the use of focus as an attentional tool. There is potentially a high processing cost to activating an alternative set and resolving it in the discourse context, so listeners may be selective. How prominent a constituent is compared to expected may be linked to how likely it is that the listener will put processing effort into generating and resolving the alternative set and intended implicatures. For example, we saw that for the false alternatives task (Yan & Calhoun, 2020, see section 5), default object bias was sufficient for listeners to process objects as focused without overt focus cues, whereas in a priming task for alternatives out of context, prosodic prominence was needed (Calhoun et al., 2025, see section 4). The processing cost of activating alternatives was greater in the latter case,

as in the false alternatives task the correct alternative was given in the preceding utterance, therefore the cues to focus needed to be stronger. Linking cues to focus with the processing of focus in this way, across languages, is likely to be important to making progress on how these work as a system.

## 7. Conclusion

In this paper we have seen how focus works as a cognitive ‘attentional tool’, to highlight particular referents in an utterance, activate alternatives and resolve intended implications. However, there are a very wide range of cues to focus, including fine phonetic detail, phonological cues, syntactic cues, semantic cues, gestural cues and contextual cues, the effects of which differ by language and by the listener and their experiences of how these fit together. I have proposed that we can think of prominence as arising from probabilistic expectations about how these cues interact. Something that is prominent is more highlighted than expected given these various cues. This works as a processing instruction to the listener, drawing cognitive attention, including to alternatives that need to be resolved and interpreted.

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## Competing interests

The author has no competing interests to declare.

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